Assign 2

과제 풀이 정리 · Notion 아카이브

Problem 3 : *Sentiment Classification*

3a - Feature(특징)추출

str -> FeatureVector

EX>

"I am what I am" --> {'I': 2, 'am': 2, 'what': 1}

==========

def extractWordFeatures(x: str) -> FeatureVector:

x_dict = {}

for y in x.split():

x_dict[y] = x_dict.get(y,0) + 1

return x_dict

3b - stochastic gradient descent

|trainExamples|, |validationExamples| : list[tuple]

|featureExtractor| : 함수

|numEpochs|, |eta|

-> WeightVector

call evaluatePredictor() -> error 출력.

output = +1 if score >= 0 else -1

score가 0일 때도 +1을 출력한다.

==========

T = TypeVar('T')

T = TypeVar('T')는 제네릭 타입을 표현하기 위한 타입 변수를 선언한다. 함수의 타입 어노테이션에서 동일한 T를 여러 번 사용하면 정적 타입

```
def learnPredictor(trainExamples: List[Tuple[T, int]], validationExamples: List[Tuple[T, int]], featureExtractor: Callable[[T], Fe
    weights = {}
    def predictor(x: T) -> int:
        score = dotProduct(weights, featureExtractor(x))
        return 1 if score >= 0 else -1
    for epoch in range(numEpochs):
```

```
    for x, y in trainExamples:
        features = featureExtractor(x)
        if dotProduct(weights, features) * y < 1:
            increment(weights, eta * y, features)
    train_error = evaluatePredictor(trainExamples, predictor)
    val_error = evaluatePredictor(validationExamples, predictor)
    print(f"Epoch {epoch}: train error = {train_error}, val error = {val_error}")
  return weights
```

Problem 4:* Multi-layer Perceptron with ComputationalGraph(그래프)*- nummpy

4a÷Node classes for computational graph

- NODE

```
class Node:
  def __init__(self, name: str, *dependencies):
    self.name = name
    self.dependencies = dependencies  # node that this node depends on (아래)
    self.value = None
    self.grad = None
  def forward(self):
    raise NotImplementedError
  def backward(self):
    raise NotImplementedError
  def grad_initialize(self):
    raise NotImplementedError


================================================


forward -> 노드 자신의 value를 찾음 (값 구하기)


backward -> dependencies의 grad를 구함
                                =dependencies에 대한 grad (파란 글씨 구하기)


grad_initialize -> 기울기 초기화(차원 맞추기)
```

- computational graph - Matrix multiplication

```
class MatMul(Node):
  """Matrix multiplication"""
```

```
def forward(self):
x, y = self.dependencies

self.value = x.value @ y.value

def backward(self):
x, y = self.dependencies

x.grad += self.grad @ y.value.T
y.grad += x.value.T @ self.grad

def grad_initialize(self):
self.grad = np.zeros_like(self.value)
```

- computational graph - Sigmoid

```
class Sigmoid(Node):
    """Sigmoid activation"""
    def forward(self):
        x, = self.dependencies  # self.dependencies가 1개의 원소를 가진 튜플일 때만 동작한다.
        self.value = 1 / (1 + np.exp(-x.value))
    def backward(self):
        x, = self.dependencies
        x.grad += self.grad * self.value * (1 - self.value)
    def grad_initialize(self):
        self.grad = np.zeros_like(self.value)
```

4b÷ MLPPredictor forward and backward

```
class MLPPredictor:
    def __init__(self, input_size, hidden_size, output_size):
        self.W1 = np.random.randn(input_size, hidden_size)
        self.W2 = np.random.randn(hidden_size, output_size)
        self.b1 = np.zeros((1, hidden_size))
        self.b2 = np.zeros((1, output_size))

# w, b 초기화.
- W1 : input_size X hidden_size
- W2 : hidden_size X output_size
- b1 : 1 X hidden_size
- b2 : 1 X output_size
```

- Y = sigmoid(sigmoid(X @ W1 + b1) @ W2 + b2)

이 모델은 이진 분류를 위한 것이며, output_size=1일 때만 Sigmoid 출력이 확률로 해석될 수 있다.

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```
 def forward(self, x):
x_node = Input("x", x.astype(np.float64))
    W1_node = Input("W1", self.W1.astype(np.float64))
    b1_node = Input("b1", self.b1.astype(np.float64))
    W2_node = Input("W2", self.W2.astype(np.float64))
    b2_node = Input("b2", self.b2.astype(np.float64))

 # 각 객체를 생성하는 단계
 z1 = MatMul("z1",x_node, W1_node)
 hidden_input = Add("hidden_input", z1, b1_node)
 hidden = Sigmoid("hidden",hidden_input)
 z2 = MatMul("z2",hidden, W2_node)
 final_input = Add("final_input", z2, b2_node)
 output_node = Sigmoid("output_node", final_input)

 #
 nodes = topological_sort(output_node)  # 노드 정렬 - numpy.ndarray
 for node in nodes:
node.forward()

self.last_nodes = nodes
self.last_output_node = output_node

return output_node.value.squeeze()  # output_node.value의 형태가 (1, 1)이면 스칼라, (B, 1)이면 (B,) 형태가 된다.
```

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```
@staticmethod  # 인스턴스나 클래스를 첫 번째 인수로 자동 전달하지 않으므로 self가 필요하지 않음
def loss(pred, target):
return (pred-target)**2  # squared loss
```

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pred : predicted probability (0 to 1), (B,)

target : true label, 0 or 1 (B,)

(B,) == 길이가 B인 1차원 배열(Vector)

output: dictionary of gradients

loss의 gradient를 구하는 것이니,

loss의 gradient는 self.last_output_node.grad = 2 * (pred_reshaped - target_reshaped)이다.

---

def backward(self, pred, target):

pred_reshaped = pred.reshape(-1,1)

target_reshaped = target.reshape(-1,1)

reshape(n, m): n은 행의 크기, m은 열의 크기

reshape(행, 열)

n=-1이면 전체 데이터의 원소 수와 m을 바탕으로 행의 크기를 자동으로 추론함.

for node in self.last_nodes:

node.grad_initialize() # 초기화

self.last_output_node.grad = 2*(pred_reshaped-target_reshaped)

for node in reversed(self.last_nodes):

node.backward()

for node in self.last_nodes:

if node.name == "W1":

W1_grad = node.grad

if node.name == "b1":

b1_grad = node.grad

if node.name == "W2":

W2_grad = node.grad

if node.name == "b2":

b2_grad = node.grad

gradients = {

"W1" : W1_grad if W1_grad is not None else np.zeros_like(self.W1),

"b1" : b1_grad if b1_grad is not None else np.zeros_like(self.b1),

"W2" : W2_grad if W2_grad is not None else np.zeros_like(self.W2),

"b2" : b2_grad if b2_grad is not None else np.zeros_like(self.b2)

}

return gradients

==========

def update(self, gradients, learning_rate):

self.W1 -= learning_rate * gradients["W1"]

self.b1 -= learning_rate * gradients["b1"]

self.W2 -= learning_rate * gradients["W2"]

self.b2 -= learning_rate * gradients["b2"]

==========

Inputs

X: input features, (N,6), N: number of samples

Y: ture labels, (N,)

epochs: number of epochs to train

learning_rate: step size for weight update

Outputs

loss: squared loss of the last step (train 함수 내부에서 계산되지만, 반환되지 않는다.)

---

def train(self, X, Y, epochs=100, learning_rate=0.1):

for i in range(epochs):

for j in range(X.shape[0]): # 행(Row)의 개수

x_sample = X[j:j+1]

y_sample = Y[j:j+1] # j번째 행

pred = self.forward(x_sample)

gradients = self.backward(pred, y_sample)

self.update(gradients, learning_rate)

loss = self.loss(pred, y_sample)

==========

def predict(self, x):

    return np.round(self.forward(x), 4)

이 함수는 예측 확률을 소수점 4자리로 반올림한 값을 반환한다. 0 또는 1로 분류하려면 0.5를 기준으로 임계값을 적용해야 하며, 현재 코드에

def predict(self, x):

    return (self.forward(x) >= 0.5).astype(int)

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